Week 1 Lecture 3

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Materials	
Module #	3
Type	Lecture
Week #	1

Client-Server & Peer-to-Peer Architecture

Architectures

- **Client-Server**
  - Server:
    - Stores data
    - Provides data on-demand
    - May also perform computations
  - Client:
    - End users
    - Requests data
    - User interaction, display
  - Network:
    - Connects the client to the server
    - Can be local
    - Data pipe - no alterations
  - **In a client-server model, the server and client are explicitly defined**
  - Local systems:
    - Both client and the server on the system ⇒ local network / communication

        - Conceptually, it is still a networked system
    - Machine clients
        - eg: Software, antivirus updaters
        - They need not have user interaction
    - Variants
        - Multiple servers, single queue, multiple queues, load balancing front-ends
    - Examples:
        - E-mail
        - Databases
        - WhatsApp / messaging
        - Web browsing
- **Distributed - Peer-to-Peer**

  Data can flow both ways
    - All peers are considered "equivalent"
        - But some peers may be more equal than others

          This just means that some servers with high bandwidth are given higher weightage
    - Error tolerance
        - Masters / Introducers
        - Election / re-selection of masters on failure
    - Shared information
    - Examples:
        - BitTorrent
        - Blockchain-based systems
        - IPFS, Tahoe (distributed file system)